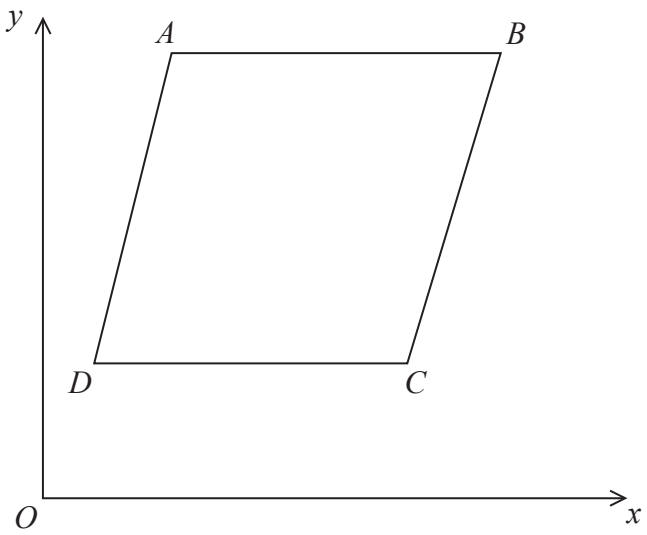
18



ABCD is a rhombus.
The coordinates of *A* are (5, 11)
The equation of the diagonal *DB* is $y = \frac{1}{2}x + 6$

Find an equation of the diagonal *AC*.

The diagonals of a rhombus are perpendicular.

DB has gradient $\frac{1}{2}$ so AC has gradient −2.

1 mark
Use perpendicular diagonals

1 mark
Negative reciprocal gradient

AC: $y = -2x + c$. Use A(5, 11).

$11 = -2 \times 5 + c$

$+10 \qquad +10$

$21 = c$

1 mark
Substitute point A

Therefore $y = -2x + 21$.

1 mark
Correct line equation

$y = -2x + 21$

(Total for Question 18 is 4 marks)